

Luiss

Libera Università Internazionale
degli Studi Sociali Guido Carli

Course Overview

Data Privacy and Security

Data Science and Management (DASMA)

LUISS



Teacher



Emilio Coppa

- Position: Assistant Professor Tenure-Track @ LUISS
- Research topics: software testing, vulnerability detection
- Other info: <https://ecoppa.github.io>
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: ecoppa@luiss.it

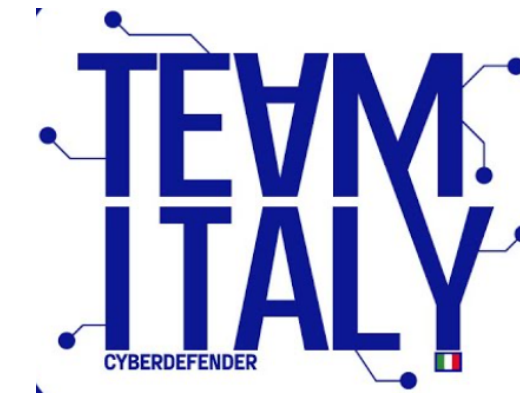
Besides research activities

CyberChallenge.IT is a national cybersecurity training program for university and high school students.



- National organizer 2017-19
- Local organizer at Sapienza 2017-23
- Sapienza has won CC.IT in 2023-25 🏆

TeamItaly is the Italian national cyber security team that takes part in institutional CTFs.



- Coach in 2017
- Team Manager in 2018-19
- Team Manager in 2024-26
- 1st position at ECSC 2025 🏆

Teaching assistants



Lorenzo Ariemma

- PhD in Computer Science and Automation
- Research topics: Computer Networks, Cybersecurity
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: lariemma@luiss.it



Lorenzo Ralli

- PhD Student in Cybersecurity @ Sapienza-LUISS
- Research topics: Software Security
- Office Hours: send me an e-mail so we can arrange a time slot.
- Contact: rallil@luiss.it

Course Goals

The course provides essential knowledge in data privacy and cybersecurity, blending theoretical concepts with practical skills:

- **Cryptography:** symmetric ciphers, asymmetric ciphers, hashing and digital signatures.
- **Authentication:** passwords, MFA, certificates.
- **Anonymous communication:** TOR and the dark web.
- **Blockchain technologies:** fundamentals and smart contracts.
- **Software vulnerabilities:** CWE, Zero day vs n-day, CVE, CVSS, OWASP Top 10.
- **Web security:** server- and client-side vulnerabilities.
- **Regulations:** Cybersecurity Act and GDPR.

When and where

1. **Monday** @ 14:45 - 16:15 (90 min)
see room schedule
(often) Theoretical lectures
2. **Tuesday** @ 15:30 - 17:00 (90 min)
see room schedule
(often) Lab Session with TA
3. **Wednesday** @ 18:15 - 19:45 (90 min)
see room schedule
(often) Lab Session with TA

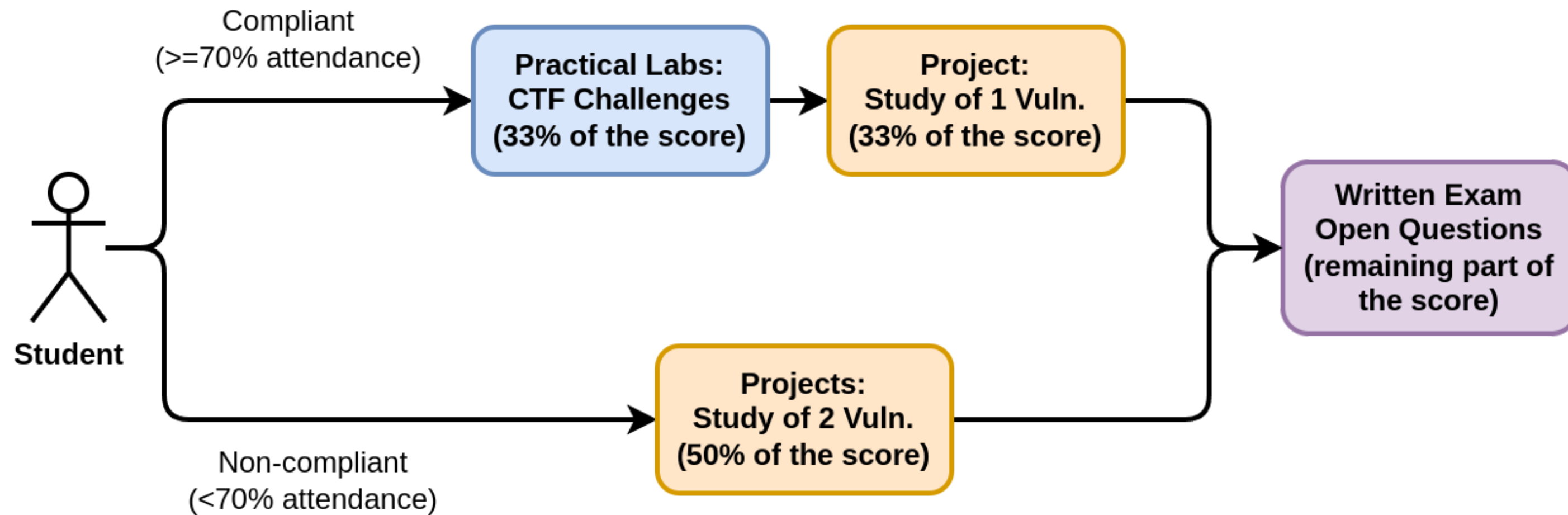
Exam

A) Attending students:

- 33% of the final grade: hands-on sessions during the course
- 33% of the final grade: project (see project description)
- 34% of the final grade: written exam

B) Non-attending students:

- 50% of the final grade: project (see project description)
- 50% of the final grade: written exam



Attending students: hands-on sessions (33% of the final score)

During the lectures, we will solve **together** some challenges:

- each **challenge** is a cybersecurity task that you have to solve
- it is inspired by a real-world scenario but with simplified conditions
- when you solve a challenge, you get a **flag** that certifies your achievement
- you have to submit the flag to our **Capture-The-Flag platform** to prove the task solution

The score is computed as the percentage of the challenges you have solved.

Attending students: project (33% of the final score)

The project:

- involves the study of **one** real-world vulnerability
- is submitted as **one** 15 pages report:
 - description of the affected program
 - description of the vulnerability type
 - description of the vulnerability
 - exploitation of the vulnerability
 - mitigation of the vulnerability
 - [optional, only for higher score] reproduction and fix of the vulnerability in a docker container

Non-attending students: project (50% of the final score)

The project:

- involves the study of **two** real-world vulnerabilities
- is submitted as **two** 15 pages report:
 - description of the affected program
 - description of the vulnerability type
 - description of the vulnerability
 - exploitation of the vulnerability
 - mitigation of the vulnerability
 - reproduction and fix of the vulnerability in a docker container

Attending and non-attending students: written exam

- It lasts one hour
- Open questions on the topics of the course
- One question may cover the topic of your project

